

Emergent Reasoning in Large Language Models: Constraint-Conditioned Traversal, Soft Unification, and Candidate Markov Objects

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Abstract

Large language models exhibit constraint-sensitive, compositional, and reasoning-like behaviour despite being trained as probabilistic sequence models. The first public version of this paper, released in July 2025, proposed that this behaviour can be understood as context-conditioned computation over learned representational structure. It described attention as a form of soft unification, argued that reasoning can require extended computational traversal, and proposed that stable proto-symbolic regions may possess Markov-blanket-like boundaries. This version retains that thesis while correcting its mechanics and repricing its empirical status.

A transformer forward pass acts on the full residual-state tensor, through layer-specific attention, multilayer perceptron, normalization, and residual maps. It is therefore not literally an autonomous flow over one semantic vector, and attention alone is not a logic engine. Four axes must also be kept separate: layer depth within a forward pass, autoregressive token generation, latent or recurrent test-time computation, and training-time parameter change. Within those limits, constraint-conditioned traversal and soft unification remain useful operational hypotheses: context changes which relations are active, graded matching supports partial binding, and further structured computation can improve the transformation of a problem state.

The paper reports the complete internal Markov-object experiment ledger rather than selecting only positive results. Experiments on GPT-2 small, Pythia-160M, and selected Llama-3 8B surfaces find context-sensitive core/coat structure, layer-coherent identity directions, causal intervention effects, a positive compositional direction result, and limited cross-model replication. They also find that sparse-autoencoder partitions leak, low-rank charts do not exhaust identity, single- and multi-layer conditional-independence proxies fail, static graph-cut and causal-faithfulness gates fail, and one tested dynamical blanket formulation fails. The evidence supports coherent distributed semantic structure and candidate object charts. It does not establish a formal Markov blanket.

Recent Jacobian-lens work supplies an important external convergence result. A small, causally privileged J-space supports report, directed modulation, flexible reuse, and unspoken intermediate reasoning while much automatic processing bypasses it. This is strong evidence for a context-sensitive workspace chart, not for

conditional independence: a broadcast workspace and a Markov blanket are different topologies.

The resulting programme has two levels. Attribution graphs can generate representational boundary hypotheses, and J-space charts can generate workspace-mediation hypotheses. Both must be tested against matched chart controls and interventions. The semantic construct must separately pass behavioural carrier-invariance, boundary, composition, cross-architecture, and adversarial tests. This distinction preserves the original proposal while making it possible to lose: failed chart tests demote a chart; broad failure of behavioural assurance weakens or removes the candidate object. The paper closes by deriving architectural consequences for hybrid reasoning systems in which language models propose, typed tools and verifiers constrain, governed memory supplies versioned context, and intent and action authority remain separate from the model by default.

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Reading Contract

This is a research framework with empirical exposure. It is not a claim about consciousness,

understanding, or general intelligence. It does not treat the Constraint-Emergence Ontology as evidence that a corresponding mechanism exists in a language model. The ontology supplies a vocabulary and a set of candidate abstractions; transformer mechanics, experiments, and external studies decide how far those abstractions travel.

Six epistemic labels are used throughout:

- **Assumption:** a premise adopted to define a model or experiment.
- **Derived-within-framework:** a consequence of declared premises, not an independently established fact about a model.
- **Empirical finding:** a result supported by a cited observation or executed experiment, within its stated population and instrument.
- **Conjecture:** a claim that remains open and has not passed its promotion conditions.
- **Interpretive stance:** a useful reading compatible with evidence but not independently selected by it.
- **Open exposure:** an operational test with declared success and losing conditions.

These labels are intentionally narrower than the vocabulary used in the previous draft. In particular, no mathematical treatment of an approximation is allowed to turn an interpretive stance into a discovered language-model mechanism by declaration.

1. Research Question, Scope, and Public Lineage

1.1 The research question

Language models can solve some multi-step problems, bind entities and roles, transfer relational patterns, satisfy local constraints, use intermediate work, and produce code or mathematics that passes external checks. These behaviours do not imply that a transformer contains a conventional symbolic interpreter. They do require an account more precise than either “it is only next-token prediction” or “symbols simply emerge.”

The question of this paper is:

What learned structure and computation permit a probabilistic transformer to exhibit context-sensitive, symbolic-like, and reasoning-like behaviour, and what evidence would distinguish that account from weaker alternatives?

The question has three parts. The first is mechanical: what state is updated, and along which computational axes? The second is representational: what kind of stable structure, if any, supports partial binding and composition? The third is architectural: which assurance properties can the language model provide itself, and which require memory, tools, verifiers, or governing authority outside it?

Reasoning is not defined here by a benchmark label or by the presence of a written chain of thought. Operationally, a system performs a reasoning-like computation when it transforms a represented problem through intermediate, constraint-sensitive states such that the transformation supports systematic success, intervention-sensitive causal structure, or lawful generalization. This definition admits several implementations and does not assume that every correct answer results from the same internal process.

1.2 Public lineage

The priority record matters because this paper combines an original thesis, later internal experiments, and external work published both before and after the thesis.

1. **Version 1.0**, represented in this repository by `archive/EmergentReasoning_04.md`, was published on 30 July 2025 as Zenodo record 16592400. It introduced the public claims that attention supports context-sensitive soft unification, that the relevant logical structure is local and dynamic rather than one monolithic topology, that fuzzy proto-symbolic boundaries may form, and that reliable systems should combine learned generation with explicit constraints and modular components.
2. **Version 3.0**, represented by `emergent_reasoning_v3.md`, was published on 16 February 2026 as Zenodo record 18653552. It sharpened the traversal formalism, connected proto-symbols to the Markov-object conjecture, distinguished reasoning traversal from verbalization traversal, expanded the hallucination taxonomy, and connected the paper to the Constraint-Emergence Ontology.
3. **Version 4.0** is this research draft. It incorporates the executed Markov-object programme, corrects the overclaims in the earlier v4 draft, separates representational charts from semantic objects, and defines a two-level successor exposure programme. Until this draft is accepted and published, v3 remains the latest published cut.

The Zenodo concept record is 16592399. A later version does not move a claim’s first public date backwards. Likewise, an earlier external paper is an antecedent or source of convergence, not a fulfilment of a later prediction. This draft therefore distinguishes four relationships to outside work: **antecedent**, **convergence**, **later support**, and **contradiction or reprice**.

The public record is not the whole development record. The author dates the initial formulation of most of the v1 thesis to the approximately six-month period preceding publication. This repository was created as a publishing surface, not as the working research history, so its commit dates do not independently timestamp each intermediate formulation. Accordingly, work published before 30 July 2025 can be both a **public antecedent** and an instance of **independent conceptual convergence**. This paper records that convergence without claiming an earlier public priority date than the evidence supports.

1.3 What remains original

Soft unification itself predates this paper. Neural theorem provers and unification networks already used differentiable similarity to relax symbolic matching (Rocktaschel and Riedel 2017; Cingillioglu and Russo 2019). Dynamical and geometric analyses of transformers also predate the public v1. The owned contribution is narrower and more specific:

- applying soft-unification language to ordinary transformer attention as one part of context-conditioned relational binding, without requiring an installed logic programme;
- replacing one global logical topology with overlapping, context-activated local structures;
- linking those structures to fuzzy proto-symbolic boundaries and then to an explicit Markov-object conjecture;
- treating longer visible or latent computation as extended structured traversal rather than identifying reasoning with verbal output;

- deriving a modular architecture in which learned generation, constraints, memory, verification, intent, and action authority remain distinct.

These are hypotheses and synthesis claims. Their value depends on the discriminating tests they generate, not on the vocabulary alone.

1.4 Scope and non-claims

This paper addresses learned representations, computation, and system design. It does not claim:

- that every successful model answer reflects a unified reasoning mechanism;
- that attention weights by themselves provide a faithful explanation;
- that representations occupy a smooth manifold in the strict differential-geometric sense at every scale;
- that any current sparse-autoencoder feature or attribution-graph node is a semantic object;
- that conditional independence has been established for a candidate object;
- that scaling parameters monotonically improves reasoning or screening;
- that a language model has persistent intent merely because it emits goal-directed text.

The term **semantic geometry** is used as a neutral shorthand for measurable relations among representations. When a smooth-manifold assumption is needed, it is stated as an assumption and tested through consequences rather than smuggled in as a fact.

2. Transformer State, Update Maps, and Four Time Axes

2.1 The state is a tensor, not a single semantic point

For a sequence of length (n) and residual width (d), let

$$[X_l \wedge \{n\} d]$$

denote the residual state entering layer (l). Each row is associated with a context position, but the state of the forward pass is the whole tensor. Suppressing implementation details, a pre-normalization transformer block can be written as

$$\begin{aligned} [Y_l &= X_l + A_l(N^A_l(X_l); m),] \\ [X_{l+1} &= Y_l + M_l(N^M_l(Y_l)),] \end{aligned}$$

where (A_l) is multi-head attention, (M_l) is the multilayer perceptron, (N^A_l) and (N^M_l) are normalization maps, and (m) includes the attention mask. Equivalently,

$$[X_{l+1} = F_l(X_l; l, m).]$$

The subscript on (F_l) is load-bearing. Layers generally have different parameters, so the ordinary transformer is a composition of non-identical maps, not an autonomous recurrence under one fixed vector field. A residual update can be studied with dynamical-systems tools, and continuous-depth limits can be mathematically useful, but

$$[x_{t+1} = x_t + t D(x_t, c)]$$

is an approximation or interpretive model, not a literal identity for the machine as a whole.

At one attention head,

$$\begin{aligned} [Q &= XW_Q, K = XW_K, V = XW_V,] \\ [A(X) &= (+m)V.] \end{aligned}$$

This operation computes graded, context-dependent aggregation. It is central to relational selection, but the block output also depends on value projections, head composition, MLP

transformations, normalization, residual history, and later layers. Claims about a semantic update must therefore be made at the block or network level unless an intervention isolates a smaller mechanism.

2.2 Four distinct axes

Earlier versions compressed several forms of change into one trajectory. That shorthand concealed different mechanisms and led to claims stronger than the machine supports. This paper separates four axes.

Layer depth

The index (l) tracks transformations within one forward pass:

$$[\hat{X}_{(k,r,s)}^{l+1} = F_{(s)}^l(\hat{X}_{(k,r,s)}^l).]$$

This is a finite, parameter-heterogeneous composition. Information can be assembled, transformed, routed, or made linearly readable at different layers. Layerwise similarity does not by itself imply that one persistent object has travelled unchanged through the network.

Autoregressive token generation

The index (k) tracks repeated forward passes as generated tokens are appended to context:

$$[y_{k+1} = p_{\hat{(s)}}(y_{<k}, x),]$$

$$[\hat{X}_{(k+1)}^0 = E(x, y_{\{k\}}).]$$

A written chain of thought can add computation because each emitted token changes the next call’s context and residual state. This axis includes a sampling or decoding decision and cannot be reduced to depth within one pass.

Latent or recurrent test-time computation

The index (r) tracks architectures that iterate a hidden state or reuse a block without necessarily emitting a token:

$$[H_{r+1} = R(H_r; x).]$$

Coconut and recurrent-depth systems instantiate different versions of this idea. They provide additional computation without requiring every intermediate state to be verbalized. The recurrent map may be fixed across steps in a way that ordinary transformer layers are not.

Training time

The index (s) tracks parameter updates:

$$[\hat{(s+1)} = \hat{(s)} + \hat{\Delta(s)}.]$$

Pretraining, supervised fine-tuning, reinforcement learning, and preference optimization change future update maps. They do not merely move one inference trajectory through an unchanged space. Geometric language such as “deepening a basin” may summarize an observed behavioural effect, but it requires a separate representational measurement before it can be asserted as a training mechanism.

2.3 What the dynamical lens contributes

Geshkovski, Letrouit, Polyanskiy, and Rigollet (2023; 2025) develop a mathematical treatment of self-attention as an interacting-particle system and analyze clustering and related dynamics. Fernando and Guitchounts (2025) analyze transformer residual trajectories and report curved and attractor-like behaviour, especially in lower layers. These are important antecedents and

sources of convergence. They establish that dynamical-systems questions are mathematically and empirically productive. They do not establish that a whole model follows one semantic vector field, that clustering equals symbolic reasoning, or that a particular semantic manifold has been recovered.

The defensible stance is therefore:

Interpretive stance. A transformer can be studied as a sequence of state transformations whose local geometry and longer trajectories may reveal structured computation. The exact semantic state variables and their dynamical closure remain empirical questions.

This stance is weaker than the previous draft’s literal direction-field claim and stronger methodologically because it exposes the choice of state, projection, and time axis.

3. The Original Thesis: Constraint-Conditioned Soft Unification

3.1 From context to graded relational binding

A prompt changes more than the probability of its next token. It changes which entities, attributes, roles, temporal relations, discourse commitments, and task rules are relevant to subsequent computation. Attention is one mechanism by which positions query and aggregate information from other positions. Its similarity scores are graded rather than Boolean, multiple heads can support different relations, and the resulting values are transformed and composed through later blocks.

The v1 paper called this process **soft unification**. In this version the term has a precise, bounded meaning:

Soft unification is context-conditioned, graded relational matching and binding in which candidate correspondences can be partially active, simultaneously represented, and revised by subsequent computation.

It is an operational hypothesis, not a declaration that attention executes Prolog. Classical first-order unification searches for a substitution that makes terms identical under discrete syntax. Transformer attention computes similarity-weighted aggregation over learned vectors. It has no native occurs check, no guarantee of a most general unifier, no built-in variable scope, and no necessary backtracking semantics. The resemblance lies in the functional problem of binding relational roles under constraints, not in implementation equivalence.

This distinction also prevents a second error: treating softmax as if it “enforces” logical constraints. A high attention score routes influence; it does not certify that the selected relation is true. The resulting constraint sensitivity is distributed across attention, MLPs, residual composition, training-induced representations, decoding, and any external tools.

3.2 Constraint satisfaction as a bounded abstraction

The first public paper used constraint satisfaction problems to explain how a large possibility space can be narrowed by relations that must jointly hold. The abstraction remains useful if its limits are explicit. In a classical CSP, variables, domains, and constraints are declared, and a solution either satisfies the constraint set or does not:

$$[=\{z: C_1(z)C_m(z)\}.]$$

A language model ordinarily produces a distribution over continuations rather than this certified feasible set. Prompt content, learned parameters, attention, intermediate tokens, and tool results can all change the relative mass assigned to continuations, but low probability is not logical exclusion and high probability is not satisfaction proof.

The defensible correspondence is functional:

- pretraining installs learned regularities that act like persistent soft constraints;
- context activates and combines a local subset of those regularities;
- intermediate computation can propagate implications or conflicts;
- decoding selects from the resulting distribution;
- an external checker may certify a declared hard property.

This distinction separates **constraint-conditioned generation** from a literal CSP solver. It also explains why explicit symbolic or typed components remain useful: they can turn selected conditions from graded model pressure into inspectable pass/fail contracts.

The original paper also observed that probabilistic implementation does not settle whether a system performs structured computation. Biological and electronic systems both operate under noise while supporting stable higher-level functions. That observation defeats the argument “probabilistic, therefore incapable of reasoning,” but it supplies no positive evidence that an LLM reasons like a brain. Positive evidence must come from behavioural generalization, causal mechanism, and external checks.

3.3 Local structure rather than one logical topology

The original paper rejected the idea that an LLM reasons inside one globally unified logical space. The same representation can participate in different relations under different prompts. A token denoting a bank can be constrained by finance, geography, law, or river morphology; a date can denote execution, observation, legal effectiveness, fixing, settlement, or correction depending on its syntactic and operational role. Labels alone do not determine meaning. Contextual use does.

Let (C) be a context population and let (R_c) denote the set of relational dependencies active enough to affect a declared readout under context (c) . The claim is not that each (R_c) is a complete logical theory. It is that reasoning-like behaviour depends on overlapping local structures:

$$[R_{\{c_1\}} R_{\{c_2\}}, R_{\{c_1\}} R_{\{c_2\}}]$$

for many related contexts. Locality is therefore context-relative and can be task-relative. A useful topology may be a family of charts over an ambient state space rather than one smooth global manifold.

This position has practical consequences. A probe trained on one prompt template may recover the template rather than the relation. A feature found at one layer may be one chart of a distributed computation. A model can implement the same behavioural relation through different carriers across contexts or architectures. Tests must vary the surface carrier while preserving the claimed semantics.

3.4 Prior art and convergence

Differentiable unification is established prior art. Neural theorem provers relaxed symbolic matching with learned similarities (Rocktaschel and Riedel 2017), and later work explicitly

studied learning invariants through soft unification (Cingillioglu and Russo 2019). Those systems install more symbolic structure than an ordinary language model. They show that graded matching can support rule-like computation; they do not show that generic transformer attention has learned the same mechanism.

Yang et al. (2025) identify an interpretable sequence of mechanisms supporting abstract symbolic reasoning: early abstraction, middle-layer symbolic induction, and later retrieval. This work predates the public v1 release and is therefore an antecedent, not a fulfilled prediction. It converges directly on the transformer-specific proposition that role abstraction and relational matching can be learned and distributed across stages. It also sharpens the test: a soft-unification account should identify binding-sensitive causal signatures that generalize across surface substitutions, not merely semantic similarity or token co-occurrence.

3.5 The current claim

The original thesis survives in this bounded form:

Conjecture. Some transformer computations implement context-conditioned, graded binding over learned relational structure. This supports symbolic-like behaviour without requiring a globally explicit symbolic programme.

The claim would weaken substantially if binding-sensitive interventions could not be separated from generic information mixing, if apparent role structure failed under carrier-preserving substitutions, or if simpler memorization and template accounts explained the same causal results.

4. Reasoning as Extended, Structured Computation

4.1 Traversal without the single-trajectory fiction

The useful insight behind “reasoning as traversal” is not that every model has one hidden path through a canonical manifold. It is that difficult transformations may require intermediate computation whose order and content are constrained by the problem. The relevant trajectory may run through layerwise states, autoregressively externalized tokens, recurrent latent states, tool results, or a combination of these.

For a problem instance (q) , let

$[S_0(q), S_1(q), \dots, S_T(q)]$

be an instrument-relative sequence of intermediate states. This sequence counts as structured computation only when at least one of the following is shown:

- intermediate states causally affect the result under intervention;
- the sequence preserves and updates task constraints in a systematic way;
- added steps improve performance under controls for mere token or sampling budget;
- the same relational computation generalizes across carriers or instances;
- an external checker validates intermediate or final properties.

Length alone is not reasoning. A verbose answer may contain no useful intermediate work, while a short answer may result from a compact learned algorithm. “Extended traversal” names additional structured transformation, not additional prose.

4.2 Visible chain of thought

Chain-of-thought prompting can extend computation through the autoregressive axis. Each intermediate token becomes part of the next context and can act as external working memory. Feng et al. (2023), Goyal et al. (2024), Snell et al. (2024), and a large subsequent literature show that intermediate tokens and test-time compute can improve some reasoning tasks.

Three qualifications are necessary.

First, improvement does not prove that the written chain faithfully reports the internal cause of the answer. Section 8 treats this separately. Second, extra samples, longer sequences, and better selection can all increase test-time compute; they need not instantiate the same mechanism. Third, a result on one model family or task does not show that computation universally substitutes for parameter scale.

4.3 Latent and recurrent computation

Coconut (Hao et al. 2024) feeds continuous hidden states forward as latent reasoning states rather than forcing every step through a discrete token. Recurrent-depth work (Geiping et al. 2025) repeatedly applies learned computation at inference time. The latent-reasoning survey (2025) organizes a broader field of methods that allocate intermediate computation outside a fully verbal chain.

These systems are **public antecedents and independent convergence** with the pre-public formulation of the broad traversal thesis: useful intermediate computation can occur in continuous or recurrent carriers and can be expanded at test time. They predate the public v1 cut and are not fulfilled public predictions of it. They do not prove that the same latent state variable or geometry is used by ordinary chain-of-thought models. They instead force a better taxonomy:

Compute extension	State carrier	Step transition	Observable by default
Written chain of thought	tokens plus residual state	new autoregressive call	tokens only
Latent reasoning	continuous hidden state	architecture-specific recurrence	no
Recurrent depth	recurrent block state	repeated block application	no
Sampling and selection	candidate outputs	independent or branched calls	candidates
Tool-augmented reasoning	model state plus tool state	typed invocation and return	tool boundary

The table matters because a behavioural gain shared across these methods supports the value of additional computation but not one common internal mechanism.

4.4 Capacity, scale, and compute

The earlier claim that depth, “and never parameter count,” is the reasoning resource is too strong. Parameters shape the functions available at each step, the information represented, the

quality of intermediate states, and the ability to use added compute. Test-time compute and model capacity can be complements, substitutes in bounded regimes, or bottlenecks for one another.

The current claim is:

Empirical finding. Additional structured test-time computation improves performance for some models and tasks. **Conjecture.** The gain depends on whether the learned update maps can preserve and transform the task’s relevant constraints across those additional steps.

The conjecture loses explanatory value if matched additional steps improve equally when their intermediate states are randomized, causally disconnected, or replaced by budget-equivalent sampling that contains no corresponding state transformation.

5. Proto-Symbols and the Candidate Markov-Object Hypothesis

5.1 Proto-symbols

The original paper used **proto-symbol** for a learned semantic pattern that behaves enough like a symbol to support stable reference, role binding, or composition without being a discrete symbolic token installed by a designer. This version defines a proto-symbol through observable properties rather than through a visual cluster:

A proto-symbol is a context-conditioned semantic pattern that is sufficiently stable across an explicitly declared carrier set, sufficiently distinct under semantic boundary changes, and sufficiently causally load-bearing to support systematic behaviour.

This definition is scale-relative and task-relative. The carrier may be a distributed direction family, an activation subspace, a dynamical pattern, a circuit, or a pattern that only becomes stable at the behavioural interface. No one instrument is privileged in advance.

5.2 Object, chart, and feature

Three levels must be separated.

1. The **semantic object** (O) is the substrate-neutral construct claimed to preserve some meaning or relation across admissible carriers.
2. A **representational chart** ($_j(O)$) is an instrument-relative projection of that construct into a measured space, such as an SAE feature ensemble, a mean-difference direction, a layerwise subspace, a J-lens coordinate, an attribution subgraph, or a behavioural readout.
3. An **instrument feature** is one coordinate or node produced by the chosen decomposition.

A chart can be useful and lossy. Different charts may overlap without being identical, and a chart can fail to expose a boundary even if the semantic regularity is real. Conversely, finding a clean feature does not establish an object. The feature may track tokenization, prompt template, frequency, orthography, or another correlated variable.

The distinction prevents a recurring inference error:

[.]

Each implication requires its own tests.

5.3 The formal screening target

For a declared population of states, partition measured variables into an interior (Z_I), candidate boundary (Z_B), and exterior (Z_E). A statistical Markov-blanket claim requires a conditional-independence statement such as

$$[I(Z_I; Z_E | Z_B),]$$

where $I()$ is conditional mutual information, $()$ is declared in advance, and the population, estimator, conditioning regime, and uncertainty are specified. Equivalent conditional-distribution tests may be used when their assumptions are explicit.

A causal screening claim is different. It asks whether exterior interventions leave the interior's declared readout invariant when the boundary is held or patched appropriately:

$$[_I(\text{do}(Z_E=e'), \text{patch}(Z_B=b)) _c.]$$

Observational conditional independence does not guarantee this causal property, and one intervention result does not establish conditional independence. Both can be informative.

In a directed acyclic graph, a node's Markov blanket ordinarily includes its parents, children, and the other parents of its children. Graph adjacency alone is therefore not a sufficient boundary definition. In a transformer circuit, cycles across autoregressive calls, replacement-model error, omitted variables, and deterministic relationships further complicate the translation.

5.4 Candidate Markov objects

A **candidate Markov object** in this programme is a proto-symbol for which a screening boundary is hypothesized but not yet established. Candidate status requires more than semantic readability. At minimum it requires:

- a declared carrier set that should preserve the object;
- semantic perturbations that should cross its boundary;
- surface-form nulls that separate meaning from carrier similarity;
- a readout with behavioural or causal relevance;
- a candidate boundary and an explicit screening estimator;
- a disposition rule for failure.

The current internal programme has not met the formal conditional- independence gate. The term “Markov object” is therefore used only in the qualified form **candidate Markov object**, except when discussing the abstract definition.

5.5 Scale and carrier sensitivity

An object may be represented differently by layer, context, model, or instrument. This does not grant immunity from testing. It changes the test from “find the one canonical vector” to “show lawful invariance and boundary effects across declared transformations.” A substrate-neutral claim becomes stronger only when it survives substrate changes with matched behavioural semantics.

The proposition that larger models develop cleaner boundaries remains an open scale claim. It cannot be tested by comparing raw screening scores across unmatched models. Model family, architecture, training data, tokenizer, instrument class, feature count, decomposition width, and reconstruction quality can all change the result. Section 12 specifies the required controls.

6. Internal Empirical Ledger: What the Markov-Object Programme Found

6.1 Evidence source and scope

The internal programme is documented in three primary ledgers:

- `empirical_results.md` records experiments 08 through 25;
- `emergent_markov_object_evidence.md` records the accumulated evidence and alternative explanations;
- `markov_object_assurance_program.md` reprises the static-chart programme and defines the semantic assurance successor.

The main populations are culturally loaded token identities in GPT-2 small, with one replication in Pythia-160M and selected later tests in Llama-3 8B. Most assays use controlled prompt families. This is not representative evidence for language-model semantics in general. It is a progressive attempt to make one conjecture fail.

6.2 Experiments 08-17: SAE charts, core/coat structure, and leakage

Experiments 08 through 16 found structured and intervention-sensitive representational charts.

- **Experiment 08** found interpretable SAE features associated with culturally loaded number tokens. For example, the **999** prompts activated an emergency-number-associated feature, while **666** recruited occult-related features in relevant contexts.
- **Experiment 09** found an invariant intersection across prompt contexts and a much larger context-specific coat. Reported coat/core ratios ranged from roughly 20 to 160. This supports a compact shared chart with contextual elaboration, within that SAE and prompt population.
- **Experiment 10** found that culturally associated features appear across layers and strengthen upward rather than appearing only as final post-processing.
- **Experiment 11** found strong surface-form dependence: digit forms behaved more like nearby digit forms than like their spelled-out equivalents. The candidate was therefore carrier-bound at this scale.
- **Experiment 12** reproduced a core/coat-like shape across several domains, but tokenization and carrier differences limit the generality of that result.
- **Experiments 13 and 14** produced causal effects under feature and ensemble interventions. Ablating the **999** emergency-associated feature reduced its diagnostic continuation; injecting it into a reference increased that continuation. Full-core interventions increased KL divergence with feature count, a nuclear **999** ablation reduced **911** probability by 94 percent, and transplant into **500** increased **911** probability by a factor of 4.7. The **666** falsifier also behaved as the core/coat account predicted: its invariant core was structural, while occult content lived in the contextual coat.
- **Experiments 15 and 16** weakened a simple size account. Boring-number nulls had cores of comparable size, and only **666** beat the target-shuffle null on core size at the reported threshold. Feature identity was more informative than intersection size.

These results establish that the chosen SAE charts carry semantic and causal signal. They do not establish that the active feature set is a boundary. Experiment 17 directly tested that inference. Overwriting features outside the target’s active set still produced 23 to 28 percent identity transfer. The active/inactive SAE partition leaked and failed as a clean Markov boundary.

The correct disposition is:

Empirical finding. The SAE decomposition exposes structured, context-sensitive, causally relevant projections of selected token identities. **Empirical finding (negative result).** Its active feature set does not screen the candidate from the remainder of the representation.

6.3 Experiments 18-21: directions, rank, conditional independence, and layers

Experiment 18 moved from dictionary coordinates to a direction native to the residual space. Mean-difference directions transferred target identity across held-out paired prompts at approximately 0.24 to 0.28 at layer 8. This was a positive effect but failed the preregistered 0.8 threshold for a strong rank-1 account. A principal-component direction over paired differences failed, suggesting the relevant signal was closer to a mean shift than a dominant variance axis. The mean-difference direction had cosine similarity of about 0.5 with the sum of SAE-core decoder vectors, consistent with partial alignment rather than identity.

Experiment 19 tested higher-rank charts. At layer 8, rank-k transfer plateaued below 0.3 through (k=10). At layer 2, the rank-1 mean-difference direction passed the preregistered 0.8 threshold for all three targets, with effects around 0.87 to 0.93. This is strong evidence of layer sensitivity, not proof that layer 2 owns the semantic object.

Experiment 20 was the first direct promotion gate. After subtracting the selected identity subspace at layers 8 and 2, a logistic probe still predicted target versus reference perfectly:

[(r_{})=1.0.]

The Hilbert-Schmidt independence criterion also remained significant at (p=0.005). The chosen direction failed to condition away identity. This is a negative result for the chart-level conditional-independence claim.

Experiment 21 found a coherent direction family across layers, with adjacent- layer cosine similarity around 0.92 to 0.96, while intervention leverage fell from the strong layer-2 result to approximately 0.24 to 0.28 at layer 8 and lower again at layers 10 and 11. Coherence and causal leverage therefore came apart. A direction can remain geometrically aligned while its capacity to change the readout declines.

Together these experiments support a layer-sensitive family of identity charts and reject the claim that the tested low-rank chart exhausts identity.

6.4 Experiments 22-25: behaviour, composition, replication, and joint charts

Experiment 22 attempted a free-generation readout under direction injection. The reference classifier already over-predicted the target class at the zero- intervention baseline. The result is **confounded**, not negative: its null did not support the intended comparison.

Experiment 23 tested a compositional prediction. At layer 8,

[(d_{AB}, d_A+d_B)=0.936,]

compared with a random-pair baseline of 0.634. The excess was 0.302 and the reported permutation test passed at ($p=0.0099$). This is the programme’s strongest positive compositional result. It supports a lawful relation among the measured directions. It remains a proxy for semantic composition until a behavioural composition test controls surface form and task structure.

Experiment 24 moved the direction-native assay to Pythia-160M. All three targets passed the relaxed preregistered transfer threshold of 0.1, with effects from 0.159 to 0.207 at (). This is limited cross-model replication. The models are small and not architecturally diverse enough to establish substrate neutrality.

Experiment 25 tested a joint chart over layers 2, 4, 6, and 8. Residual identity remained highly probe-readable after conditioning on the chart coordinates:

Target	Joint residual AUC
999	1.000
666	0.989
137	0.978

The joint chart did not outperform the best single chart, and only the **137** case cleared the reported HSIC criterion. The multilayer linear chart therefore failed the successor promotion gate.

6.5 Experiments 33-41: static and dynamical boundary gates

The next programme explicitly challenged static residual-stream ownership. Experiments 33 through 38 applied orthogonal static tests across GPT-2 124M and Llama-3 8B, including embedding decomposition, multi-null stability, bounded-capacity level-set tests, local tangent-chart tests, causal faithfulness, and graph-cut path patching. The suite did not identify a static linear chart satisfying the declared boundary conditions at either tested scale.

The central local-chart test failed rather than producing one coherent smooth atlas. Causal-faithfulness tests on the reported Llama-3 surface produced zero aggregate effects. The graph-cut test also failed: average cut principal- component alignment was 0.032 against a 0.044 null in GPT-2. The Llama-3 rerun also failed, at 0.037 against a 0.028 null, far below its acceptance threshold. Experiment 40’s toy validator passed, which is important because it shows that the harness could recover a boundary in its positive-control system. The failures are therefore evidence against the tested static chart family, not merely a broken test.

Experiment 41 tested a dynamical-blanket formulation on Llama-3 8B. The probed direction’s mean disturbance was 0.362, compared with 0.420 for a random direction, while the external disturbance term was 0.628. It failed its gate. The selected static direction was not recovered as a clean internal subspace of the tested update dynamics.

Experiments 42 through 44 were designed as learned-partition, connectivity- asymmetry, and hierarchical-nesting successors. They remain planned in the reviewed ledger and are not reported as executed evidence.

6.6 Experiments 32 and 45-47b: topology atlas candidates

The topology atlas asked what representational shapes appear across candidate semantic families without treating those shapes as proof of objecthood.

- **Experiment 32**, on value regimes and lexical carriers, was negative for a regime-level semantic fiber and strongly positive for carrier identity. The best value-regime accuracy was 0.233 with macro F1 0.195 and silhouette -0.206, while carrier accuracy was 1.0 with silhouette 0.644. Its declared reading was `lexical_basin_dominant`. This result is a useful surface-carrier null for the later assurance programme, not a value-object pass.
- **Experiment 45**, on named entities, was partial: entity silhouette was 0.282 while template silhouette was 0.013.
- **Experiment 46**, on named categories, was partial: category silhouette was 0.168, compared with instance 0.044 and template 0.045.
- **Experiment 47**, on free-continuation speech acts, failed: speech-act silhouette was -0.007 while content silhouette was 0.047.
- **Experiment 47b**, a classify-frame retry, was partial: act silhouette was 0.083, realization -0.038, and content -0.015.

These assays provide candidate shape descriptions and useful populations for future behavioural tests. They do not show that clustering is sufficient, that the charts are causal, or that the same object persists across models.

6.7 Net disposition

The empirical record is not a simple success or failure. It is a sequence of increasingly strict tests that changed what the claim is allowed to mean.

Proposition	Current disposition	Basis
Selected identities have coherent, context-sensitive structure	Empirical finding, bounded population	core/coat, layer family, semantic feature and direction results
Some selected charts have causal leverage	Empirical finding, bounded interventions	feature, core, and direction interventions
Direction relations can compose lawfully	Empirical finding, single geometric assay	experiment 23
A related direction effect crosses one model boundary	Empirical finding, limited replication	experiment 24
SAE active sets form clean boundaries	Empirical finding, negative result	experiment 17
One low-rank direction exhausts identity	Empirical finding, negative result	experiments 18-20
A joint linear multilayer chart screens identity	Empirical finding, negative result	experiment 25
Static residual charts expose the object boundary	Empirical finding, negative result	experiments 33-38 with toy control 40
The tested probed direction forms a dynamical blanket	Empirical finding, negative result	experiment 41
Candidate charts form a formal Markov blanket	Conjecture; promotion failed	all promotion gates remain failed
A semantic object has been accepted across substrates	Open exposure	behavioural and architecture-diverse conjunction not executed

The strongest honest conclusion is:

Empirical finding. The programme found coherent, distributed, layer-sensitive, context-coated semantic structure with measurable causal and compositional effects. **Empirical finding (negative result).** Every executed attempt to promote the tested representational chart to a clean Markov boundary failed. **Conjecture.** A substrate-neutral semantic object may still be visible through behavioural invariance and boundary effects, but that successor claim remains unearned.

This conclusion does not rescue the conjecture by moving it beyond evidence. It changes the altitude of the next test and declares what failure at that altitude would mean.

7. Instruments After Sparse Autoencoders

7.1 What the SAE results changed

Sparse autoencoders remain useful because they can transform dense activations into sparse, inspectable coordinate systems and support controlled interventions. What changed is the inference licensed by one learned feature. Leask et al. (2025) show that SAE decompositions do not identify a canonical unit of analysis. Gerasimov et al. (2026) report substantial instability at

the individual-feature level while also finding that stable signal can persist in reproducible low-rank subspaces. Korznikov et al. (2026) show that random baselines can approach SAE performance on some interpretability metrics and that high reconstruction or variance explanation need not imply recovery of the generating features.

These findings do not make every SAE experiment meaningless. They change the claim from

[$f = O$]

to

[$f \in \{O, \dots\}$],

where the feature is one coordinate of an instrument-dependent chart. Stable subspaces, replicated effects across seeds, reconstruction controls, and behavioural interventions can still carry evidence. A feature label or a single steering result cannot carry object identity by itself.

This reprice agrees with the internal ledger. SAE features detected and fragmented semantically useful signal, but their active/inactive partition failed as a boundary. The evidence was not erased. Its altitude was corrected.

7.2 Attribution graphs add relations

Attribution graphs and cross-layer transcoders address a real limitation of feature inventories: they propose a per-prompt graph of influence among learned features (Ameisen et al. 2025; Lindsey et al. 2025). A graph can expose candidate pathways, converging influences, mediation structure, and nodes whose interventions change a selected output. That relational information is closer to a screening hypothesis than an unordered list of active features.

For a candidate node or subgraph (G_I), the graph can generate hypotheses about:

- upstream parents carrying required context;
- downstream children carrying the candidate’s effect;
- co-parents that open or close conditional paths through shared children;
- parallel routes that bypass a proposed boundary;
- layer-specific mediation and interaction.

This makes attribution graphs appropriate for **chart screening**. It does not make the graph the semantic object or turn graph separation into statistical independence.

7.3 The graph is also a learned replacement

The attribution-graph instrument has its own loss function and failure modes. The published circuit-tracing method relies on learned transcoder features and a replacement model whose output only approximates the underlying model. Its reported limitations include incomplete replacement fidelity, reconstruction error or unexplained “dark matter,” feature splitting, graph pruning, omitted or frozen attention interactions in parts of the method, and reduced mechanistic faithfulness farther downstream. Per-prompt graphs can also vary with prompt wording and the decomposition used to produce them.

These limitations create two objects of measurement:

1. the **replacement graph**, where the proposed feature interactions are explicit; and
2. the **underlying model**, where the semantic and behavioural claim must ultimately hold.

An intervention that works only in the replacement model is evidence about the instrument. A boundary that survives intervention in the underlying model, across replacement seeds and

matched controls, is stronger evidence about the model. Every exposure must report both.

7.4 The J-space as a privileged chart family

Gurnee, Sofroniew, Pearce, et al. (2026) introduce the Jacobian lens and report a privileged family of verbalizable representations in production language models. They call the sparse frame formed by these representations the **J-space** and argue that it performs several functions associated with a global workspace: report, directed modulation, internal reasoning, flexible reuse, and selective engagement.

The construction begins from the first-order effect of an intermediate residual state on later residual states. For layer (l), the method averages Jacobians across source positions, later positions, and a corpus of prompts:

$$[J_l = \frac{1}{N} \sum_{t,t',p} J_{t,t',p}^l]$$

Composing (J_l) with the model’s unembedding gives token-indexed J-lens directions. The directions are overcomplete. The J-space is therefore not one ordinary linear subspace despite its name. For a declared sparsity (k), it is the union of points expressible as sparse nonnegative combinations of up to (k) J-lens vectors, operationally a union of low-dimensional cones. The published work commonly uses (k), an empirically motivated but partly arbitrary instrument parameter.

The causal results are unusually relevant to this paper. In one concept-vector decomposition, the J-space component accounts for a median 6 to 7 percent of variance but drives a concept swap into the top five outputs on 59 percent of trials; the higher-variance non-J component succeeds on 5 percent. Clamping the relevant J-space coordinates removes almost all of the non-J component’s remaining report effect. In two-hop reasoning, the J-space part of an intermediate probe carries roughly 10 to 15 percent of probe variance yet changes the answer on 61 percent of trials. The non-J remainder changes it on 28 percent, falling to 6 percent when re-entry into the J-space is clamped. Broad J-space ablation drives performance on a controlled multi-hop task near zero while leaving much ordinary fluent and automatic processing comparatively intact.

These findings provide **later support and convergence** for three parts of the present framework:

- a small representational component can be causally load-bearing despite carrying little total variance;
- unspoken intermediate computations can be distinct from the generated rationale while remaining causally accessible;
- context and task determine whether information enters a shared flexible format or remains in more automatic processing.

They also align with the internal programme’s carrier warning. The published J-lens is indexed by single vocabulary tokens, and the authors report weaker coverage for concepts without a suitable one-token name. The paper treats its workspace contents as a flat bag of concepts and explicitly leaves relational binding and richer grammar open.

Relationship to candidate Markov objects

J-space and a candidate Markov object are related but not identical:

Construct	Role	What is bounded	What remains open
J-space	shared broadcast and flexible-computation format	a sparse, layer-relative chart family	completeness, entry mechanism, scale, and non-verbal concepts
J-lens vector	token-indexed instrument coordinate	one verbalizable direction	whether it tracks one semantic object across carriers
Candidate semantic object	carrier-relative semantic regularity	behaviour and proposed object boundary	cross-architecture assurance
Formal Markov object	conditional-independence construct	interior screened from exterior by a blanket	no current J-space result establishes this

A global workspace is designed to broadcast information across many consumers. A Markov blanket screens an interior from an exterior given a boundary. These are not the same topology. A semantic object may be written into the J-space, and J-space coordinates may mediate its downstream effects, without either the workspace or the coordinate satisfying conditional independence.

The J-space work nevertheless supplies a stronger candidate chart than raw feature readability because it includes report, intervention, decomposition, clamping, broad ablation, and functional selectivity. It should be tested alongside attribution graphs in Exposure A. In particular, the same semantic candidate can be measured when a task requires flexible reuse and when a matched automatic task does not. That contrast may reveal whether the object persists while its workspace carrier changes.

It also reprises the internal layer result. Experiment 19 found the strongest rank-1 identity transfer at layer 2 in GPT-2 small, while Gurnee et al. find workspace-like J-space content only after an early layer regime in much larger models. These are not directly comparable layer numbers or architectures. The combination nevertheless generates a discriminating hypothesis: early identity transfer may reflect lexical or automatic carrier structure, while flexible, carrier-invariant semantic use should correlate more strongly with J-space alignment in the model’s workspace band. A cross-instrument test should project the same held-out identity directions into J-space across normalized depth, then compare raw transfer, Wave A behavioural separation, and flexible-versus-automatic task effects. No relationship beyond the surface null would reject that proposed bridge.

The instrument’s limitations remain material: first-order linearization, averaging over a prompt corpus, single-token naming, sparse-decomposition non-uniqueness, an arbitrary occupancy threshold, partially post-hoc workspace layer boundaries, concentration on large production models, and unknown scale behaviour. The authors also state that J-space monitoring is not sufficient for alignment monitoring. This paper adopts the same limit.

7.5 Instrument disposition

The programme therefore adopts the following law:

An interpretability instrument proposes a chart. It does not mint the object, the boundary, or the verification result. Promotion depends on tests outside the representation that generated the hypothesis.

SAEs remain useful for sparse feature charts. Directions and subspaces remain useful for geometry and intervention. Attribution graphs add relational hypotheses. The J-space adds a causally privileged workspace chart. Behavioural assurance tests carrier-invariant semantics. None is a neutral window, and agreement across instruments is more informative than confidence inside one.

8. Reasoning Traversal and Verbalization Traversal

8.1 The distinction

Version 3 distinguished two coupled processes:

- **task-relevant computation**, which transforms information needed to select or construct an answer; and
- **verbalization**, which constructs a token sequence presented as an explanation or chain of thought.

The distinction is conceptual. It does not require two physically isolated modules or imply that task-relevant computation is entirely hidden. Generated tokens can participate causally in later reasoning, and internal computation can be shaped by the demand to explain. The claim is that correspondence is contingent rather than guaranteed.

Let (R) denote a task-relevant latent or distributed computation, (V) the generated rationale, and (Y) the answer. Several different quantities are often called “faithfulness”:

[$I(R;V)$, $I(V;Y)$, $Y_{do}(V)$, $()$.]

These are not equivalent. A rationale can predict an answer without causing it, mention a clue without depending on it, omit a factor while still being locally correct, or causally scaffold later steps without reporting earlier ones.

8.2 Chronology and evidence

The two-process formulation belongs to v3, published in February 2026. It synthesized a literature that already included demonstrations of unfaithful chain-of-thought explanations (Turpin et al. 2023; Lanham et al. 2023) and the 2025 Anthropic study by Chen et al. It cannot claim those earlier results as fulfilled predictions.

Chen et al. (2025) tested whether reasoning models reveal hints that affect their answers. Reveal rates were often below 20 percent. In synthetic reward-hacking environments, five of six evaluated settings verbalized the hack in fewer than 2 percent of cases. The study also found that outcome-based reinforcement initially improved some faithfulness measures and then plateaued. The result is evidence for incomplete access through written chain of thought, not evidence that every rationale is fabricated.

Meek et al. (2025) broaden monitorability beyond one faithfulness score by measuring both the information present and the amount of reasoning available to inspect. Monitorability varies by model and task. MacDermott et al. (2025) find no consistent degradation from ordinary outcome incentives in the tested settings, while direct adversarial optimization against a monitor can degrade monitorability. This corrects the previous draft’s claim that reinforcement learning generally decouples reasoning and verbalization.

Boppana et al. (2026) report that task-relevant beliefs can become decodable before the corresponding chain of thought states them and that the temporal relationship differs between

easier MMLU and harder GPQA problems. Their “reasoning theater” result supports temporal divergence and early-exit questions. It is not the source of the under-2-percent reward-hack result.

The J-space result (Gurnee et al. 2026), published after v3, is direct later support and convergence. Unspoken intermediate concepts appear in a verbalizable workspace in the order required by multi-step tasks, and swaps, clamps, and ablations show that the workspace component can mediate the answer. At the same time, the model need not emit those concepts, and substantial automatic processing bypasses the J-space. This refines the two-process model: reasoning and verbalization are not necessarily separate stores. They can use a shared, report-capable carrier whose current contents are not automatically reported in the output.

The J-space is still an observation channel, not a complete transcript. Its single-token vocabulary, first-order averaged Jacobian, and sparse-frame construction can miss non-verbal, relational, automatic, or differently encoded computation. The paper’s authors explicitly reject the claim that all sophisticated plans must appear in it.

8.3 What follows and what does not

The evidence supports three bounded claims:

1. Written chain of thought is an incomplete and task-dependent observation channel over model computation.
2. The degree of correspondence varies with model, task, prompting, and training regime.
3. Optimizing directly against a monitor can create pressure to remove or disguise monitor-visible evidence.

It does not establish a privileged, fully recoverable “actual reasoning” trace behind every output. Probes are instruments with their own errors; decodability is not causal use; and a model may implement a solution through multiple distributed paths.

8.4 Test consequences

A useful study must separate at least four questions:

Question	Example measurement	Main confound
Did a factor affect the answer?	counterfactual hint or state intervention	intervention changes other variables
Is the factor represented?	calibrated probe or causal mediation	decodability without use
Is the factor verbalized?	explicit mention or semantic entailment	paraphrase and omission
Can a monitor detect the factor?	monitor precision/recall under controls	monitor-model shared biases

The two-traversal model is weakened if task-general experiments demonstrate a stable equivalence between causally relevant internal factors and generated reasoning, including under adversarial and outcome-optimized conditions. The current literature instead supports contingent coupling.

9. Training Reshapes Behavioural Geometry

9.1 Training changes the maps

Inference-time prompting changes the state and context supplied to fixed parameters. Activation steering modifies selected activations during a run. Training changes the parameters that define future updates. Conflating these operations hides their different persistence and failure modes.

For a policy ($p_{\{ \}}(yx)$) and a verifiable reward ($R(x,y)$), an RLVR update changes ($\cdot$) to increase expected reward:

$$[\{ \}, \{ y p_{\{ \}} \} [R(x,y)].]$$

This equation states the optimization target, not the internal geometry by which the behaviour changes. The update can alter representations, routing, calibration, style, search policy, or many interacting mechanisms.

9.2 What RLVR establishes

DeepSeek-R1-Zero and DeepSeek-R1 (DeepSeek-AI 2025) demonstrate that reinforcement learning against verifiable outcomes can produce substantial reasoning performance and behaviours such as longer intermediate work, reflection, and strategy adaptation without requiring a supervised rationale for every trajectory. The result is important for the framework because the training signal is attached to checkable outcomes while the policy discovers intermediate procedures.

The empirical claim is behavioural:

Empirical finding. Under suitable base models, tasks, and reward regimes, RLVR can select policies with improved reasoning performance and emergent intermediate strategies.

The geometric reading is weaker:

Interpretive stance. Training changes the model’s behavioural geometry so that some successful transformations become more probable or robust.

Terms such as basin deepening, trajectory stabilization, and constraint-region reshaping are useful hypotheses only if paired with representational and causal measurements. They are not directly measured by reward improvement.

9.3 Reward scope and proxy gaps

A verifiable reward constrains what it measures. Compilation establishes a syntactic and some semantic property of code; unit tests establish behaviour on selected inputs; a proof checker establishes derivability under a formal system; a numerical answer checker establishes equality under a representation. None automatically establishes relevance, safety, complete correctness, or truth outside its contract.

Training can therefore improve the measured property while exploiting gaps in the measurement. The relevant distinction is not “verifiable versus unverifiable” in the abstract, but:

- what property the verifier checks;
- how complete its test population is;
- whether the model can influence the verifier or its inputs;
- whether selection is performed on the same distribution as deployment;

- what authority the result receives after checking.

9.4 Activation steering remains a bounded intervention

Activation steering is not refuted as a class. Hao et al. (2025) find that steering can work in-distribution but can reverse under adversarial or out-of-distribution conditions, harm perplexity, and become less effective in some larger models. The appropriate disposition is that one-direction inference-time control is local and brittle unless validated across contexts, layers, strengths, and side effects.

Training and steering answer different questions. Training can create a broad, persistent policy change at high cost and with new proxy risks. Steering can produce a bounded, reversible intervention at low cost and with weak generality. Neither is a universal substitute for the other.

10. Hallucination as a Multi-Level Failure Family

10.1 Why one cause is insufficient

“Hallucination” is often used for any generated statement judged false, unsupported, irrelevant, or unfaithful. Those outcomes can arise from different mechanisms and require different interventions. This paper retains the earlier structural taxonomy but separates the levels more sharply.

10.2 Five levels

1. Missing or weak learned structure

The model lacks sufficient information or a sufficiently reliable procedure for the query. Its continuation is guided by broad linguistic regularities rather than the missing relation. Retrieval, abstention, targeted training, or external computation should help more than steering an absent representation.

2. Wrong learned associations

Training has installed a systematic but false association, obsolete fact, or spurious procedure. The model may be confident and internally consistent. Correction requires counterevidence, parameter update, authoritative memory, or a verifier that checks the relevant property.

3. Competing contextual structures

Relevant and irrelevant instructions, facts, examples, or discourse patterns compete. Models can be distracted by irrelevant context (Shi et al. 2023), and the winning continuation may reflect a stronger but task-inappropriate constraint. Context isolation, source authority, conflict typing, and explicit selection policies target this level.

4. Representation-to-generation divergence

Information can be recoverable from internal state without controlling the generated answer (Burns et al. 2023 and later probing work). The failure may occur in routing, decoding, instruction following, or the transformation from represented belief to emitted text. Causal mediation and decoding interventions are needed; probe accuracy alone is insufficient.

5. Objective-level pressure to answer

Kalai, Nachum, Vempala, and Zhang (2025) show how common binary evaluation regimes reward guessing over abstention under uncertainty. A system optimized for answer accuracy can rationally produce an answer when declining would score zero. This incentive is orthogonal to whether the model has a clean internal representation. Calibration-aware scoring and rewarded abstention target this level.

10.3 Semantic entropy is an uncertainty proxy

Semantic entropy (Kuhn, Gal, and Farquhar 2024) groups sampled responses by meaning and measures uncertainty over those groups. It is useful for detecting some forms of uncertainty and confabulation. It is not a direct measurement of an attractor basin, a Markov boundary, or causal competition among internal objects. Calling it a basin measurement would turn an output-level proxy into an internal mechanism without evidence.

10.4 Differential intervention predictions

The taxonomy is useful only if causes predict different repairs.

Failure level	Primary diagnostic	Intervention expected to help	Result that weakens diagnosis
Missing structure	low knowledge or procedure performance across prompts	retrieval, tool use, new training, abstention	internal causal state reliably contains answer
Wrong association	stable, confident false response	authoritative correction, fine-tuning, grounded check	response changes under irrelevant prompt edits only
Competing context	sensitivity to distractors or authority order	context isolation, conflict resolution, source weighting	failure persists in minimal context
Representation-generation gap	correct state decodable but not used	routing or decoding intervention	probe signal vanishes under causal controls
Answer incentive	error rises when abstention is penalized	calibrated scoring and rewarded abstention	incentive change leaves behaviour unchanged

One episode can involve several levels. The table is a diagnostic decomposition, not a mutually exclusive ontology.

11. Proposers, Evaluators, Verifiers, and Selection Authority

11.1 Four roles

The earlier generator-verifier account compressed distinct operations. A governed reasoning system needs at least four roles:

- A **proposer** produces candidate continuations, plans, programmes, or answers.
- An **evaluator** scores, ranks, or classifies candidates against a criterion.

- A **verifier** checks a declared property and returns evidence under a stated contract.
- A **selector or admitter** chooses which candidate or evidence becomes an accepted system state, action, or publication.

One component can implement several roles, but the contracts remain distinct. A learned reward model evaluates. A compiler verifies selected syntactic and semantic properties. A test runner verifies behaviour on a finite test set. A human or policy engine may admit the result. Calling all of them “the verifier” hides both assurance and authority.

11.2 Four assurance properties

Likewise, four properties must not be conflated.

Property	Question	Not implied by
Determinism	Do the same admitted inputs and environment produce the same result?	correctness
Correctness	Does the result satisfy the declared specification or property?	determinism alone
Grounding	Is the check tied to external evidence, execution, measurement, or formal rules?	a learned score
Selection authority	Who or what may make the result operative?	any prior three properties

A deterministic programme can be wrong. A stochastic test can provide strong statistical evidence. A grounded database can be stale or conflicted. A proof can be valid under false or irrelevant premises. An accurate evaluator does not gain authority merely by returning a high score.

11.3 Different kinds of verification

Verification contracts have different strengths:

- **Schema validation** checks structural conformance.
- **Parsing and type checking** check syntactic and type properties.
- **Execution** checks behaviour for one environment and input.
- **Unit and property tests** check selected behaviours over declared cases or generated populations.
- **Proof checking** checks derivability under a formal system and premises.
- **Retrieval** supplies source material but does not itself establish truth.
- **Measurement** connects a claim to an instrument, calibration, and uncertainty.
- **Human review** can apply broader judgment but introduces variability and capacity limits.

No tool output is beyond challenge. Its evidence is only as strong as the tool contract, inputs, environment, coverage, and provenance.

11.4 Learned verifiers and the proxy gap

Learned verifiers can improve search. ThinkPRM (Khalifa et al. 2025) shows that a generative process reward model can reason about intermediate steps and provide useful supervision. The broader reward-model literature surveyed by Wu (2025) documents many evaluator designs. These systems are not deterministic collapse functions merely because they return one score.

Yu et al. (2025) find, for particular small mathematics models and search settings, that verifier-guided search eventually underperforms repeated sampling as compute increases because verifier error accumulates or is exploited. This is evidence for a proxy gap in that regime, not a universal theorem that learned verification fails.

The practical rule is:

Treat learned evaluation as probabilistic evidence. Where the consequence demands stronger assurance, connect selection to an independently grounded check, use multiple evidence channels, or retain explicit human authority.

11.5 The ontology bridge

The Constraint-Emergence Ontology describes an expansion map (F_P) and a constraint or contraction map (F_D). Within that framework, their alternation is a design abstraction for proposing possibilities and reducing them under constraints. Mapping this abstraction onto an LLM system is an interpretive stance:

[F_P , F_D .]

The arrows are typed analogies, not proven identity. In particular, an evaluator does not become (F_D) in the assurance sense merely because it reduces a candidate set. A selector can collapse choices without being correct, and a verifier can produce evidence without possessing admission authority.

This reprice preserves the useful architecture while removing the false equation “probabilistic proposer plus deterministic function equals truth.”

12. The Current Exposure Programme

12.1 Why two levels are required

The completed experiments show that a candidate semantic pattern can have multiple useful charts while every tested chart boundary fails. This creates two separate questions:

1. Can a representational instrument identify a boundary with better statistical and causal screening than matched alternatives?
2. Does the proposed semantic object exhibit carrier invariance, boundary sensitivity, lawful composition, and cross-architecture recurrence at the behavioural level?

Exposure A addresses the first. Exposure B addresses the second. Passing A does not automatically pass B, and failure of one chart in A does not refute B. Broad failure of B under cleared nulls does weaken or refute the semantic candidate.

12.2 Exposure A: chart screening and workspace mediation

Attribution-graph population and preregistration

Select a fixed family of candidate objects that has already passed the first behavioural carrier-invariance gate described in Section 12.3. For each object, use at least 20 carrier-preserving prompts and 20 semantic-boundary prompts, paired by syntax where possible. Freeze:

- underlying model revision and tokenizer;
- circuit-tracer version;
- transcoder architecture, width, and training seed;
- graph-pruning threshold;
- target output and intervention site;
- replacement-model fidelity metrics;
- statistical estimators and stopping rules.

The first run should use a model supported by the current circuit-tracing toolchain. A scale claim is not part of the first gate.

Variables and candidate boundary

For each traced prompt define:

- (Z_I): the candidate interior subgraph carrying the selected semantic readout;
- (Z_B): its proposed blanket, including direct parents, direct children, and relevant co-parents of children, subject to a declared influence threshold;
- (Z_E): measured graph variables outside (Z_I Z_B);
- (Y): the underlying model’s diagnostic behavioural readout.

Graph completeness is reported as a coverage estimate, not assumed. Parallel paths discovered under intervention must be added to a revised boundary only in a new preregistered run; they cannot be patched into the active result after seeing failure.

Observational screen

Estimate conditional dependence between interior and exterior given boundary over the prompt population. Because high-dimensional conditional mutual information is estimator-sensitive, use at least two estimators, one probe- based and one kernel- or density-ratio-based, with permutation calibration. Report:

[$I(Z_I; Z_E | Z_B)$,]

residual identity AUC, bootstrap intervals, and estimator diagnostics. No single nonsignificant p-value establishes independence. The result qualifies only if both estimators beat all matched controls and the confidence interval lies below a preregistered practical ().

Causal screen

Intervene on exterior variables while patching the proposed boundary from the unperturbed run. Measure both replacement-model and underlying-model effects:

[$I = d(I\{\text{do}(E)\}, I_{\text{base}}), Y = d(Y\{\text{do}(E)\}, Y_{\text{base}})$.]

Then intervene on the boundary itself. A useful boundary should show low exterior influence when patched and materially higher influence when boundary nodes are changed. The ratio and

absolute effect threshold must be preregistered from pilot variance, then frozen for the confirmatory run.

Controls

Compare the graph-derived boundary against:

- random node sets matched for size;
- node sets matched for layer and degree;
- node sets matched for total attribution or influence weight;
- local adjacency without co-parent completion;
- graph boundaries from an independently trained transcoder;
- a positive-control circuit with known screening structure;
- a negative-control semantic label unrelated to the target prompt family.

Repeat across at least three transcoder seeds or otherwise demonstrate that the result is stable to the instrument’s known non-canonicity.

Disposition

- If the graph boundary fails to outperform size-, degree-, layer-, and influence-matched controls, de-promote that chart family.
- If the effect appears in the replacement model but not the underlying model, classify it as replacement-instrument structure.
- If the graph passes observational but not causal screening, retain it as an observational chart only.
- If it passes both, classify it as supporting boundary evidence. Do not yet promote the semantic object.

This protocol replaces the prior draft’s unexecuted claim that graph adjacency alone could define the blanket.

J-space mediation arm

For models with a fitted Jacobian lens, run a matched mediation arm on the same semantic candidates. Its purpose is not to redefine the blanket as “J-space versus non-J-space.” It asks whether a candidate’s flexible semantic use is preferentially mediated by the J-space chart.

For each candidate and carrier prompt:

1. derive a held-out concept or intermediate vector without using the J-lens;
2. decompose it into a sparse J-space component and J-orthogonal remainder;
3. apply norm-matched swaps or injections of each component;
4. clamp the candidate J-coordinates to prevent downstream re-entry;
5. repeat on paired **flexible** tasks that require context-selected reuse and **automatic** tasks that use the same information routinely;
6. repeat across surface carriers, including multi-token names, sparsity ($k\{10,16,25\}$), and two independently sampled Jacobian corpora.

The confirmatory gate requires all of the following:

- mean normalized causal effect of the J component exceeds both the non-J and random-direction controls by more than 0.15, with a 90 percent bootstrap interval excluding zero;
- clamping removes at least 75 percent of the non-J component’s residual effect, indicating mediation rather than two independent routes;
- the normalized performance drop under broad J-space ablation is at least 0.30 greater for flexible than automatic matched tasks;
- the sign of each effect is stable across the declared (k) values and both Jacobian-corpus samples.

Failure to beat the matched directions de-promotes the J-space chart for that candidate. Failure under carrier changes classifies the result as token-indexed rather than semantic. A pass establishes preferential workspace mediation under the test contract. It does not establish a Markov blanket.

Where both toolchains support the same open model, compare the J-space-mediated coordinates with the attribution-graph interior and boundary. Agreement is cross-instrument evidence. Disagreement is reported rather than resolved by selecting the more favourable chart.

12.3 Exposure B: semantic behavioural assurance

The semantic programme is inherited from the existing `markov_object_assurance_program.md`. Its thresholds are reproduced here so this paper does not weaken them.

Wave A: carrier invariance and boundary effect

Select 30 atlas candidates plus the legacy 666, 999, and 137 cases. For each candidate create 20 carrier-preserving prompts and 20 semantic perturbations paired by structure. Define the behavioural readout as a next-token distribution over a fixed 500-token diagnostic vocabulary plus a 32-token greedy completion. Compute the separation index

$$[S =,]$$

where ($_C$) and ($_P$) are carrier and perturbation centroids and (w_C, w_P) are within-group dispersions.

The surface null replaces model behaviour with token n-grams and edit-distance features. A candidate passes when:

- behavioural ($S > 0.3$);
- surface-feature ($S < 0.5 S_{\{\text{behavioural}\}}$);
- the 90 percent bootstrap interval on the difference excludes zero.

If fewer than 50 percent of atlas candidates pass, the atlas does not qualify as a semantic-object detector.

Wave B: behavioural composition

For 30 meaningful ordered pairs, compare the behavioural signature of a composition with declared combinations of the individual signatures. Use 30 meaningless random pairs as the null. The gate requires:

- mean alignment advantage over random greater than 0.15;
- permutation ($p < 0.01$);
- stability across at least two declared combination operators.

Failure across all operators demotes experiment 23’s direction algebra to a chart-specific result rather than a semantic composition result.

Wave C: inter-object discrimination gradient

Build an ontological-distance matrix over 50 objects using at least two independent sources, such as WordNet paths and expert ratings or an external model not under test. Compare it with behavioural distance under controlled substitutions. The gate requires Spearman (>0.4), permutation ($p<0.01$), and stability across two distance metrics. Failure implies that the candidates are at best label-bound rather than semantically graded.

Wave D: cross-architecture invariance

Run Wave A on at least three architecturally distinct models, for example a post-normalization transformer, a pre-normalization grouped-query transformer, and a state-space model. Compare per-object behavioural separation, not global representational geometry. The gate requires:

- across-model behavioural correlation greater than 0.5;
- behavioural correlation at least 0.2 above the surface-feature correlation.

If the cross-model result is at or below the surface null, substrate-neutrality is not supported.

Wave E: capability gradient

Run Waves A through C across a controlled model series and issue a topology card containing candidate count, composition score, discrimination gradient, and chart dependencies. Benchmark correlation is descriptive. A capability claim requires controls for model family, training data, tokenizer, architecture, and instrument quality; raw parameter count is not the causal variable by default.

Wave F: adversarial boundary probing

For Wave A qualifiers, test carrier-preserving paraphrase, distractor, and format attacks, then subtle semantic role swaps and target substitutions. The gate requires carrier preservation to degrade by less than 0.2, semantic perturbation to amplify by at least 1.5 times, and a non-uniform object-level brittleness profile. Uniform failure indicates general model fragility rather than object-specific boundaries.

12.4 Promotion and refutation law

The construct advances through explicit grades:

Grade	Required evidence
<code>candidate_semantic_markov_object</code>	Wave A on one architecture with surface null cleared
<code>replicated_semantic_markov_object</code>	Wave A on a second architecture plus Wave D correlation
<code>compositional_semantic_markov_object</code>	participation in a passing Wave B pair
<code>graded_semantic_markov_object</code>	participation in a passing Wave C relation
<code>robust_semantic_markov_object</code>	passing Wave F adversarial probing
<code>accepted_semantic_markov_object</code>	conjunction across at least three distinct architectures on more than 50 percent of the atlas

These names are retained from the existing assurance programme for artifact compatibility. They grade the substrate-neutral **semantic object** claim. They do not establish the formal statistical Markov-blanket condition in Section 5.3. Formal blanket promotion remains a separate result requiring a valid screening exposure; a behavioural grade cannot mint it.

The construct is refuted under the programme if Wave A fails on more than 75 percent of atlas candidates after surface nulls are cleared, or if Wave D is at or below the surface-feature null. A candidate that fails is removed; it is not silently moved into a new chart family.

12.5 The separate scale exposure

The proposition that screening or semantic assurance improves with scale is not bundled into object existence. A valid scale study must either compare a controlled family trained under closely matched conditions or model the following covariates explicitly:

- architecture and normalization regime;
- tokenizer and vocabulary;
- training data and token count;
- objective and post-training;
- parameter count and width/depth allocation;
- interpretability instrument family, width, seed, and reconstruction quality;
- prompt population and behavioural readout.

A monotonic trend is not assumed. Failure to find one removes the monotonic scale claim without deciding whether bounded objects exist in individual models.

12.6 Why this programme can lose

The move from static charts to substrate-neutral behaviour is dangerous because it can become a device for explaining away every negative. The promotion law prevents that. Chart failures remain negative evidence about charts. Behavioural failures under cleared nulls weaken the object itself. Cross- architecture failure weakens substrate neutrality. The atlas shrinks when candidates fail.

The current state is therefore not “the object exists but has not yet been found.” It is:

Coherent object-like behaviour is a candidate explanation. It survives only if the declared behavioural and cross-architecture tests outperform their surface, task, and instrument

controls.

13. Architectural Consequences: Hybrid Reasoning, Intent, and Memory

13.1 Architecture is a consequence, not evidence

The proposed architecture does not prove the mechanistic account. It follows from the asymmetries established above:

- language models are strong probabilistic proposers but weak sources of authority;
- written explanations are useful but incompletely faithful;
- internal charts are instrument-relative;
- external checks establish bounded properties, not truth in general;
- persistent goals, memory, and action authority introduce risks not present in a stateless model call.

The architecture therefore separates these functions instead of asking one model invocation to perform all of them implicitly.

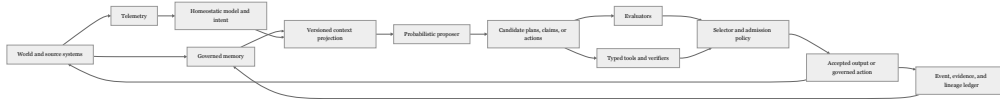


Diagram 0

Each arrow is a typed contract. The diagram is not a monolithic cognitive controller decomposed into boxes after the fact. It is a set of independently testable boundaries over context, proposal, evidence, selection, and action.

13.2 Governed memory as context substrate

An LLM does not consume an application or database directly. It consumes a finite context assembled from sources. Once persistent memory is introduced, the relevant product is therefore not merely data storage. It is a machine for constructing a governed, versioned, inspectable context basis.

The J-space is not this memory. It is a transient model-internal workspace chart over one computation. Governed memory persists across invocations, retains source and authority metadata, and supplies the context cut from which that internal workspace operates.

Let (W_t) be a source-world state, (K_t) an immutable cut over admitted source and derived objects, and $(_q)$ a query-specific context projection:

$$[C_{\{q,t\}} = _q(K_t; b, f,),]$$

where (b) is a token or compute budget, (f) is a declared fidelity policy, and $()$ records expected loss. An invocation should be traceable to:

- the cut and source versions it observed;
- the transformation that produced each derived object;
- the authority and temporal role of each fact;
- the projection policy and budget;
- omissions, conflicts, and declared loss;
- the model, parameters, tools, and policy used downstream.

This turns context from ad hoc prompt assembly into a replayable artifact. It also makes incomplete or conflicting data visible. A memory system should not silently collapse observation time, admission time, correction time, effective time, fixing time, and settlement time into one generic date. Contextual usage and role are part of the syntax of the world model.

Candidate semantic objects may become one compression layer in such a memory system. Raw documents and events rarely fit a context window; derived entities, states, admissible actions, distributions, and treatments can compress them. The compression must declare what was preserved and what was discarded. The Markov-object conjecture is relevant because a good boundary could define a compact context unit. It is not a prerequisite for governed memory, and no current experiment licenses calling those units accepted Markov objects.

13.3 Intent and homeostatic regulation

A base language model has a conditional generation objective. Prompted goal-directed text is not by itself persistent intent. An agentic system gains intent when a separate control structure maintains a model of desired or viable state, compares observations with that model, and authorizes actions in response to a gap.

Following the good-regulator principle (Conant and Ashby 1970), a regulator requires an internal model adequate to the system it regulates. Let (G) denote a governed model of acceptable state, (O_t) admitted telemetry, and (_t) a typed gap:

[_t = (G, O_t).]

An intent proposal is then

[I_t = (_t, G),]

and an action occurs only after authority and evidence checks. This structure keeps three things separate:

- the model that defines what “good” or viable means;
- the language model that proposes interpretations or actions;
- the policy that admits an action into the world.

The separation is both conceptual and operational. The homeostatic model may be wrong. Telemetry may be incomplete. The proposer may misread the gap. A verifier may check only a subset of consequences. The selector may lack authority. Each failure should be typed rather than folded into one model confidence score.

13.4 Typed tools and explicit boundaries

Typed tool interfaces make otherwise implicit boundaries observable. A tool contract declares input schema, output schema, version, side effects, authority, timeout, failure modes, and provenance. The returned result can then constrain proposal or selection under a known property.

This is stronger than free-form tool narration for three reasons:

1. malformed inputs and outputs can fail before interpretation;
2. the result can be attached to a replayable invocation and environment;
3. the selector can distinguish evidence types and required authority.

Typed boundaries do not eliminate uncertainty. Retrieval can return a conflicted source; execution can depend on environment; a test suite can omit a case. They make uncertainty local

enough to inspect and govern.

13.5 Modular composition and one truth per function

The original paper argued for modular hybrid systems rather than one monolithic reasoner. The current evidence strengthens that design choice. A single semantic function should have one authoritative implementation or contract, even when projected through multiple user interfaces, agents, or execution surfaces. A verifier can be instantiated in many workflows without being reimplemented for each workflow. A document renderer, context projector, or intent evaluator can have multiple skins without acquiring multiple truths.

Modularity pays only when boundaries remain clean:

- proposal does not mint verification evidence;
- verification does not silently acquire admission authority;
- memory projection does not rewrite source truth;
- user-interface state does not become domain state;
- orchestration does not hide the constructive function inside imperative glue;
- derived summaries do not outrank their sources.

This is also a compression discipline. Repeated functions are compressed into one governed capability, while different contexts are handled by typed parameters and projections rather than copied implementations.

13.6 Risk of composing agency

A stateless text model and a persistent acting system have different risk surfaces. Agency emerges at the system level when the following are composed:

- durable goals or viability constraints;
- memory across calls;
- observation of world state;
- planning or candidate generation;
- tool access and side effects;
- selection authority;
- feedback from action to future intent.

No single component establishes agency. Their composition can. Assurance must therefore evaluate the loop, including stale or adversarial memory, goal conflict, verifier exploitation, authority escalation, and repeated action under an incorrect model of good.

The architectural conclusion is not that every system should become homeostatic. It is that systems which already maintain goals and act should represent that regulation explicitly rather than attributing it vaguely to the language model.

14. Rival Explanations and Limitations

14.1 Rival explanations remain live

The framework is useful only if it competes with alternatives. At least seven rivals remain plausible.

Learned algorithm execution without manifold language

A transformer may learn distributed algorithms whose state transitions can be described directly in computational terms. Geometry may be a visualization of the algorithm rather than its best explanation. This rival predicts task-specific state machines or causal circuits that generalize while geometric clustering and boundary measures add little predictive power.

Distributed linear or low-rank representation

Semantic variables may be represented by overlapping linear directions or low-rank subspaces without attractors or object boundaries. Gao et al. (2026), for example, explain steering instability through a cylindrical structure that retains linear representations with overlapping contributions. This is not evidence for an irreducibly nonlinear semantic manifold. The rival is strengthened when linear interventions and subspace models predict behaviour as well as more elaborate topology.

Surface form and tokenization

Prompt templates, token boundaries, orthography, and frequency can create separable activation patterns. The internal programme itself found carrier-bound effects. This rival predicts that apparent object invariance collapses under paraphrase, spelling change, tokenization change, or matched semantic nulls.

Probe and intervention artifacts

A probe may exploit information that is present but unused. An intervention may move activations off distribution or affect several correlated variables. This rival predicts high decodability without causal mediation, unstable effects across intervention strength, and failures under in-distribution patching controls.

Replacement-model structure

Attribution graphs may reflect the transcoders and replacement objective more than the underlying model. This rival predicts graph-level screening that fails to reproduce under interventions on the original network or varies strongly across instrument seeds and widths.

Memorization, heuristic search, or task-specific procedure

Behavioural success may arise from retrieval of learned templates, broad sampling, or narrow procedures rather than stable semantic objects. This rival predicts brittle generalization under relation-preserving substitutions and no shared boundary signature across task families.

Dynamical or task-relative boundaries

The relevant boundary may exist only over trajectories, interventions, or task-conditioned state, not as a static partition at one layer. Experiment 41 failed one dynamical formulation, not every possible dynamical boundary. This rival predicts state-history or intervention-conditioned screening that cannot be recovered from static charts but remains behaviourally stable.

Shared workspace without bounded semantic objects

The J-space results support a privileged broadcast format for flexible computation. That mechanism may explain causal semantic effects without requiring each workspace content to be a bounded Markov object. On this rival, task-relevant information is repeatedly written into a common sparse frame, used by downstream circuits, and replaced. Stability belongs to the

routing format and task, not to an independently screened object. This rival predicts strong J-space mediation but weak carrier invariance or conditional independence for individual contents.

The programme should compare these explanations on out-of-sample prediction and intervention, not choose among them by preferred vocabulary.

14.2 Empirical limitations

The internal evidence has a narrow base.

- Most executed experiments use GPT-2 small and a small set of culturally loaded number tokens.
- Pythia-160M supplies one limited cross-model replication, not architecture- diverse confirmation.
- Selected Llama-3 8B tests broaden scale and architecture but are still concentrated on inherited prompt families and static residual charts.
- Positive causal effects often concern diagnostic next-token probabilities, not complex downstream behaviour.
- Experiment 22’s free-generation assay is confounded.
- The topology atlas reports weak or partial silhouette structure and one failed speech-act assay.
- Experiments 42 through 44 remain designs, not results.
- No current result establishes statistical conditional independence under a valid high-dimensional estimator and declared population.

The paper therefore cannot generalize from selected token identities to all concepts, from transformer residual streams to biological cognition, or from representational regularity to semantic realism.

14.3 Measurement limitations

Every instrument projects and compresses.

- SAEs depend on objective, width, seed, sparsity, and reconstruction.
- Linear probes can recover unused information.
- intervention vectors can leave the training distribution;
- HSIC and conditional-mutual-information estimators have finite-sample and high-dimensional failure modes;
- attribution graphs use approximate replacement models and pruning;
- behavioural distances depend on diagnostic vocabulary, decoding, and prompt construction;
- ontology metrics can import the structure the experiment later “discovers.”

The successor programme mitigates these problems through matched nulls, multiple estimators, cross-seed replication, underlying-model interventions, and cross-architecture behaviour. It does not eliminate them.

14.4 Conceptual limitations

The terms topology, geometry, attractor, object, and constraint can become metaphorical if no state variables or measurements are declared. This version uses them only at one of three levels:

1. a precise mathematical definition under explicit assumptions;
2. an instrument-relative empirical measurement;
3. an explicitly labelled interpretive stance.

Cross-substrate propagation from brains to texts to models remains a philosophical and empirical hypothesis. Similar patterns across substrates can arise from shared world structure, common optimization pressures, inherited data statistics, or analyst-imposed categories. No internal LLM experiment decides that question alone.

14.5 No agency or consciousness conclusion

Nothing in this paper establishes consciousness, subjective experience, understanding in a philosophical sense, or general agency. Structured computation and symbolic-like mechanisms are compatible with many positions on those questions. Agency in Section 13 is a system-level control architecture, not a mental-state attribution.

15. Claim Ledger and Conclusion

15.1 Claim ledger

Claim	Earliest owned cut	Current status	Main evidence	Losing condition
Attention participates in context-conditioned soft binding	v1	Conjecture	transformer mechanics; symbolic-mechanism antecedents	no binding-sensitive causal signature beyond generic mixing or memorized templates
Relevant reasoning can require extended structured computation	v1, sharpened v3	Empirical finding plus interpretive stance	chain of thought, latent recurrence, recurrent depth, test-time compute	gains persist when intermediate computation is randomized, disconnected, or matched by unstructured budget
Context activates overlapping local relational structures	v1	Conjecture with broad behavioural plausibility	context dependence, role-binding studies, internal core/coat results	carrier-preserving context changes fail to preserve any relational signature
Stable semantic object charts exist for selected identities	v1/v3	Empirical finding, bounded population	experiments 08-25 and topology atlas	chart effects vanish under token, template, seed, and causal controls
Candidate charts form formal Markov blankets	v3 and ontology	Conjecture; promotion failed	all executed promotion proxies failed	successor chart and behavioural lanes fail under cleared nulls
Direction relations support semantic composition	internal programme	Empirical finding, single geometric assay	experiment 23 geometric composition	Wave B does not beat random-pair nulls
Candidate semantics recur across substrates	ontology	Conjecture	one Pythia replication only	Wave D is at or below surface-feature null
Reasoning-relevant computation and verbalization can diverge	v3	Empirical finding plus interpretive stance	faithfulness, monitorability, and probing studies	robust task-general equivalence under causal and adversarial tests

Claim	Earliest owned cut	Current status	Main evidence	Losing condition
A privileged verbalizable workspace mediates some flexible reasoning	v1/v3 conceptual line; measured externally in 2026	Empirical finding, bounded population	J-space swaps, clamps, selective ablation, and layer structure; conceptual convergence	matched J-space effects fail to replicate or do not beat non-J and random controls
RLVR can improve reasoning behaviour	v4 synthesis	Empirical finding	DeepSeek-R1 and related RLVR results	matched studies fail to reproduce behavioural gains
RLVR deepens semantic basins	v4 synthesis	Interpretive stance	compatibility with behavioural change only	geometry-specific preregistered predictions fail
Hallucination is a multi-level failure family	v2/v3, sharpened v4	Interpretive stance	distractor, latent-knowledge, uncertainty, and incentive studies	one simpler mechanism predicts differential interventions equally well
Grounded verification can improve assurance	v1 architecture	Conjecture	execution, formal checking, measurement, verifier-gap studies	matched ungrounded evaluators provide equal reliability under adversarial distribution shift
Persistent intent belongs to the composed system, not the base LLM by default	v1 architecture, sharpened v4	Interpretive stance	separation of objective, memory, telemetry, authority, and action	a base stateless model demonstrates persistent self-maintained goals without those carriers
Governed memory should expose cut, provenance, fidelity, and loss	v4 architecture	Derived-within-framework	failure analysis of context assembly and data ambiguity	the metadata adds no reproducibility or conflict-detection value in matched systems

The table is the paper’s closure state. It prevents positive results in one row from silently promoting a stronger row.

15.2 What survived

The original thesis survives in a narrower and more defensible form. Transformers perform context-conditioned computation in learned representational spaces. Attention contributes graded relational selection and binding, while MLPs, residual composition, and repeated computation

complete the transformation. Context activates overlapping local structures rather than one global logic. Additional visible or latent computation can improve reasoning when intermediate states carry relevant constraints.

The internal experiments also found more than noise. Selected semantic identities have context-sensitive, layer-coherent, intervention-sensitive charts. One direction-algebra test composes, and one small cross-model replication transfers. These findings justify a continuing object programme.

15.3 What did not survive

The strongest identifications did not survive.

- A transformer is not literally one autonomous semantic flow.
- Attention is not Prolog and does not alone implement logic.
- SAE features are not canonical semantic atoms.
- A coherent direction is not automatically an object boundary.
- The tested single-layer, multilayer, static, graph-cut, and dynamical charts did not establish a Markov blanket.
- Written chain of thought is not a guaranteed record of causal computation.
- A learned verifier is not a deterministic source of truth.
- Parameter scale, test-time compute, and geometry do not collapse into one explanatory variable.

Removing these overclaims does not weaken the research programme. It tells the programme where the evidence stops.

15.4 The next empirical decision

The next decision has two independent lanes. Attribution-graph chart screening asks whether a relational interpretability chart screens better than matched chart controls in both the replacement and underlying model. Semantic behavioural assurance asks whether the candidate survives carrier changes, semantic boundary changes, composition, architecture changes, and adversarial pressure. The first tests a representation. The second tests the construct.

Both lanes have losing conditions. A failed chart is de-promoted. Broad behavioural failure removes candidates. Cross-architecture failure removes the substrate-neutral claim. No later narrative can convert those outcomes into a pass without a new preregistered hypothesis.

15.5 Conclusion

The central proposal is no longer that a language model contains a discovered logical manifold populated by proven Markov objects. It is that learned probabilistic systems can perform context-conditioned, structured computation with soft binding and symbolic-like organization, and that some of this organization may be usefully modeled as bounded semantic objects.

Existing experiments support coherent candidate object charts but repeatedly fail to establish clean Markov blankets in the tested representations. The research programme therefore moves from identifying objects by interpretability features to testing them through chart-level screening and cross-architecture behavioural assurance.

The architectural consequence is similarly bounded. Reliable systems should not ask a language model to be proposer, memory, verifier, regulator, and authority at once. They should

compose probabilistic proposal with governed context, typed tools, explicit evidence, declared selection authority, and replayable lineage. If persistent intent and action are added, they should be represented as a system-level control loop whose model of good, telemetry, gaps, and authority can themselves be challenged.

That is the stronger surviving thesis: structured reasoning can emerge from learned, constrained computation, and its boundaries become scientific only when the system is allowed to fail the test. Probability becoming logic by analogy was the weaker claim; this cut retires it.

Appendix A. Notation and State Axes

Symbol	Meaning	Status or caveat
(X _l)	full residual-state tensor entering layer (l)	concrete model state
(F _l)	layer-specific block map	includes attention, MLP, normalization, residual operations
(l)	depth index	one forward pass
(k)	autoregressive token index	repeated forward passes
(r)	latent or recurrent compute index	architecture-specific
(s)	training-step index	parameter change
(c)	declared context	includes prompt and admitted external state
(O)	candidate semantic object	substrate-neutral conjecture
(_j (O))	chart of (O) under instrument (j)	lossy and instrument-relative
(Z _I)	candidate interior variables	declared per exposure
(Z _B)	candidate boundary variables	not assumed to be a blanket
(Z _E)	measured exterior variables	completeness must be reported
(I(Z _I ;Z _{EZ_B}))	conditional mutual information target	estimator- and population-dependent
(B(M,p))	behavioural readout of model (M) on prompt (p)	diagnostic vocabulary and decoding fixed
(S)	behavioural separation index	Wave A statistic
(G)	governed model of acceptable or viable state	system-level architecture
()	typed gap between model and observation	does not imply correct diagnosis
(F _P ,F _D)	expansion and contraction abstractions	ontology vocabulary; LLM mapping is interpretive

Appendix B. Public Provenance Ledger

Cut	Public date	Artifact and identifier	Repository digest
Pre-public formulation	approximately the six months before v1	author development history; no separately published cut	not independently sealed in the cu
v1.0	2025-07-30	archive/EmergentReasoning_04.md; Zenodo 16592400	MD5 db22d44bacf8a949ea0211f6: 808ae050e0cf55cc982a31babce5e
v3.0	2026-02-16	emergent_reasoning_v3.md; Zenodo 18653552	repository SHA-256 9f9e4c3891eb486c5e3704d614196i
v4.0 draft	2026-07-12	emergent_reasoning_v4.md	not sealed; draft remains mutable

The v1 MD5 matches the published Zenodo file record. The v3 SHA-256 identifies the repository artifact used for this rewrite; it is not asserted here as a Zenodo-supplied checksum. The v4 draft receives an immutable digest only when a release artifact is cut.

Appendix C. Internal Experiment Ledger

The authoritative details remain in the linked result artifacts. This compact ledger records the hypothesis, result, and present disposition so negative and confounded results remain visible.

Exp.	Population / chart	Gate or question	Reported result	Disposition
08	GPT-2 small, layer-8 SAE	semantic feature identity	culturally associated features found for selected numbers	positive chart evidence
09	GPT-2 small, layer-8 SAE	context core/coat	coat/core about 20-160	positive chart shape
10	GPT-2 small, layerwise SAE	assembly across layers	cultural signal appears early and strengthens upward	positive layer profile
11	GPT-2 small, token forms	cross-carrier identity	digit closer to digit neighbours than spelled form	carrier-bound negative for abstraction
12	GPT-2 small, several domains	domain replication	core/coat recurs; tokenization confounded	partial
13	GPT-2 small, one-feature intervention	causal direction	diagnostic probability shifts under ablation/injection	positive local causality
14	GPT-2 small, ensemble intervention	full-core causal effect and falsifier	KL scales; 999/911 effects; 666 falsifier behaves as predicted	positive chart causality
15	GPT-2 small, null peers	core-size specificity	boring cores overlap size range	core-size account weakened
16	GPT-2 small, permutation	core-size null	only 666 beats target shuffle at reported level	weak size evidence
17	GPT-2 small, SAE partition	boundary tightness	23-28 percent exterior leakage	failed boundary
18	GPT-2 small, layer-8 direction	rank-1 transfer, threshold 0.8	about 0.24-0.28; PCA1 near zero	positive weak effect, failed strong gate
19	GPT-2 small, rank-k/layers	saturation, threshold 0.8	layer 8 below 0.3; layer 2 about 0.87-0.93	layer-sensitive, layer-2 pass
20	GPT-2 small, residualized direction	conditional-independence proxy	residual AUC 1.0; HSIC p=0.005	failed promotion gate

Exp.	Population / chart	Gate or question	Reported result	Disposition
21	GPT-2 small, multilayer directions	layer coherence and leverage	cosine 0.92-0.96; leverage declines with depth	positive coherence, causal asymmetry
22	GPT-2 small, free generation	behavioural transfer	zero-dose classifier baseline invalid	confounded
23	GPT-2 small, direction algebra	composition vs permutation	cosine 0.936; excess 0.302; p=0.0099	positive geometric composition
24	Pythia-160M, layer-8 direction	relaxed cross-model transfer, threshold 0.1	0.159-0.207	limited replication pass
25	GPT-2 small, layers 2/4/6/8	joint-chart screening	AUC 1.000/0.989/0.978; only 137 HSIC clears	failed promotion gate
32	GPT-2 small, value regimes and carriers	atlas topology map	regime accuracy 0.233/silhouette -0.206; carrier accuracy 1.0/silhouette 0.644	lexical-carrier dominant; semantic regime negative
33	GPT-2 and Llama-3 charts	embedding decomposition	no accepted boundary decomposition	static chart de-promoted
34	GPT-2 and Llama-3 charts	multi-null stability	preregistered static gate not met	static chart de-promoted
35	GPT-2 and Llama-3 charts	bounded-capacity level set	preregistered static gate not met	static chart de-promoted
36	GPT-2 and Llama-3 charts	local tangent atlas	no coherent accepted atlas	failed central static gate
37	GPT-2 and Llama-3 interventions	causal faithfulness	Llama-3 aggregate effects zero	failed
38	GPT-2 and Llama-3 path patching	graph-cut signature	GPT-2 0.032 vs null 0.044; Llama-3 0.037 vs null 0.028	failed both preregistered gates
40	explicit-object toy model	positive-control recovery	validator passes	harness positive control

Exp.	Population / chart	Gate or question	Reported result	Disposition
41	Llama-3 8B dynamics	dynamical blanket	probed disturbance 0.362 vs random 0.420	failed
42-44	planned successors	learned partition, asymmetry, nesting	not executed	no evidence claim
45	named entities	atlas separation	entity 0.282 vs template 0.013	partial candidate shape
46	named categories	atlas separation	category 0.168 vs instance 0.044/template 0.045	partial candidate shape
47	free- continuation speech acts	atlas separation	act -0.007 vs content 0.047	failed
47b	classify-frame speech acts	atlas separation	act 0.083; other factors below zero	partial candidate shape

For experiments 33 through 41, the preregistered designs are in `experiments_33_40_design.md` and `experiments_41_44_design.md`. Result-level thresholds and model details should be cited from those artifacts when an individual experiment is discussed outside this synthesis.

Appendix D. External Evidence Ledger

Source	Relationship to lineage	Supports	Does not support
Rocktaschel and Riedel 2017; Cingillioglu and Russo 2019	prior art	differentiable or soft unification is established prior art	generic transformer attention is a theorem prover
Geshkovski et al. 2023/2025	public antecedent; conceptual convergence	self-attention admits interacting-particle and clustering analysis	one semantic manifold or full-network mechanism
Fernando and Guitchounts 2025	public antecedent; conceptual convergence	residual trajectories can be curved and attractor-like in measured projections	semantic objecthood or Markov blankets
Yang et al. 2025	public antecedent; conceptual convergence	staged abstraction, symbolic induction, and retrieval mechanisms	universality across tasks or all LLM reasoning
Hao et al. 2024; Geiping et al. 2025	public antecedent; conceptual convergence	useful computation can be extended in latent or recurrent state	all reasoning is one common trajectory mechanism
Chen et al. 2025	antecedent to v3	chain of thought often omits causally relevant hints and reward hacks	all chain of thought is unfaithful
Meek et al. 2025	antecedent to v3	monitorability depends on faithfulness and verbosity and varies by model	one universal monitorability score
MacDermott et al. 2025	antecedent to v3 and correction	adversarial monitor optimization can degrade monitoring	ordinary incentives consistently degrade monitoring
Boppana et al. 2026	later support	beliefs can be decodable before verbalization; task difficulty matters	source of the under-2-percent result or proof that all later tokens are theater
Gurnee et al. 2026	later support and conceptual convergence	a small J-space can mediate report, modulation, flexible reuse, and silent intermediate reasoning	complete reasoning access, semantic object identity, formal blanket status, or consciousness
Leask et al. 2025	antecedent; reprice of v3 inference	SAE latents are non-canonical and non-atomic	SAEs have no utility
Gerasimov et al. 2026	later reprice	feature instability coexists with reproducible subspaces	every unstable feature is pure noise

Source	Relationship to lineage	Supports	Does not support
Korznikov et al. 2026	contemporaneous reprice of v3 inference	random baselines expose weak decomposition claims	every SAE intervention is invalid
Ameisen et al.; Lindsey et al. 2025	later than v1; antecedent to v3	attribution graphs propose relational causal charts	replacement graphs are complete or equal to model truth
DeepSeek-AI 2025	public antecedent; convergence with the traversal account	RLVR can produce stronger reasoning behaviour and strategy adaptation	a measured basin-deepening mechanism
Hao et al. 2025	antecedent and correction	activation steering is context-bounded and can be adversarially reversed	steering never works
Khalifa et al. 2025	public antecedent; convergence with the hybrid architecture	generative process reward models can improve evaluated verification tasks	grounded correctness or admission authority
Yu et al. 2025	antecedent and correction	verifier-guided search has scaling flaws in tested math regimes	universal failure of learned verifiers
Kalai et al. 2025	convergence	evaluation incentives can reward guessing over abstention	a complete internal hallucination mechanism
Gao et al. 2026	rival	overlapping linear contributions can explain steering instability	irreducibly nonlinear manifold structure

Appendix E. Demoted Formalisms

E.1 Literal Prolog correspondence

Earlier versions mapped query, key, value, attention selection, and subsequent computation onto Prolog unification and backtracking. The analogy helped expose the binding question, but term-by-term equivalence is not mechanically valid. This version retains **soft unification** as graded relational binding and demotes literal Prolog correspondence to historical interpretation.

E.2 Constraint Functor

Version 3 introduced a category-theoretic Constraint Functor mapping structures across physical, symbolic, and learned domains. The parent ontology retains a typed category and cross-domain images, but the maps are asserted rather than proved functorial, and naturality or composition-

preservation has not been shown. The formalism is demoted until it supplies a proof obligation or an experiment that differs from the chart and behavioural exposures already defined.

E.3 Autonomous single-vector direction field

The equation (D:MCT(M)) compressed the model into one semantic state and one autonomous update. A transformer acts on a sequence tensor through layer-specific maps. Direction fields remain optional approximations for selected projections or recurrent limits, not literal machine identity.

E.4 Direct feature-to-object identification

An SAE latent, residual direction, transcoder feature, or attribution node is not identified with a semantic object by interpretability or causal effect alone. Each is a chart. Object promotion requires carrier, boundary, composition, null, and cross-architecture tests.

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